Pinboard Studio - Project 2 Execution Plan

Issue dependencies, recommended team order, owner split, Google Photos API pivot, and rubric grading map

> **Preserved boundary: the execution-plan scope/order through Phase 3 Issue #3 is retained. Issue #3 remains closed and its GitHub issue definition was not edited. Phase 1, Phase 2, and #3 GitHub issue contents were left unchanged. This revision updates later work to Google Photos.**

How to use this plan

Complete dependencies in order. Within a phase, work may run in parallel only when no dependency remains open. GitHub status labels below reflect the repository when this PDF was generated; they are not a claim that every earlier phase is fully complete. For example, issue #4 remains open and is unchanged.

Rubric note: the supplied rubric states 335 possible points, but some text is truncated or poorly rendered, especially the video and Vision/Gemini point ranges. Verify the final Canvas allocation. This map preserves the original criteria and points while describing the approved Google Photos pivot.

Google Photos integration rules

Use Google Photos Picker for the user's own selected photos. The user searches/selects in Google's Picker UI; the app must not enumerate the whole library. Import selected bytes promptly because Picker base URLs are temporary, validate them through issue #1, and store the private working copy in the existing Cloud Storage lifecycle from #3. Firebase login is separate from Google Photos OAuth. The Picker scope is `photospicker.mediaitems.readonly`. Optional export (#26) uses the Library API only to create new app-generated media; it never edits the selected original and does not block the core release.

Recommended execution schedule

Phase 1 - Foundation

#2 - Project foundation: replace scaffold with production-ready Express structure [CLOSED / COMPLETE]

Owner: RyanTorrs | Area: Foundation

Dependencies: None

github.com/bhushannakhate/project-2/issues/2

#22 - Configure Firebase project, environment variables, and service account access [CLOSED / COMPLETE]

Owner: RyanTorrs | Area: Foundation

Dependencies: Project foundation: replace scaffold with production-ready Express structure

github.com/bhushannakhate/project-2/issues/22

#14 - Configure CI, linting, formatting, and pull-request checks [CLOSED / COMPLETE]

Owner: bhushannakhate | Area: Foundation

Dependencies: Project foundation: replace scaffold with production-ready Express structure

github.com/bhushannakhate/project-2/issues/14

Phase 2 - Identity and data

#16 - Implement Firebase Authentication API and protected routes [CLOSED / COMPLETE]

Owner: RyanTorrs | Area: Authentication

Dependencies: Configure Firebase project, environment variables, and service account access

github.com/bhushannakhate/project-2/issues/16

#21 - Define Firestore collections, indexes, security rules, and data helpers [CLOSED / COMPLETE]

Owner: bhushannakhate | Area: Database

Dependencies: Configure Firebase project, environment variables, and service account access

github.com/bhushannakhate/project-2/issues/21

#11 - Build the React sign-in and account experience [CLOSED / COMPLETE]

Owner: bhushannakhate | Area: Authentication UI

Dependencies: Configure Firebase project, environment variables, and service account access, Implement Firebase Authentication API and protected routes

github.com/bhushannakhate/project-2/issues/11

#4 - Implement authenticated user profile API and credits [OPEN]

Owner: bhushannakhate | Area: Credits

Dependencies: Implement Firebase Authentication API and protected routes, Define Firestore collections, indexes, security rules, and data helpers

github.com/bhushannakhate/project-2/issues/4

#17 - Implement creation CRUD API [CLOSED / COMPLETE]

Owner: bhushannakhate | Area: Database

Dependencies: Implement Firebase Authentication API and protected routes, Define Firestore collections, indexes, security rules, and data helpers

github.com/bhushannakhate/project-2/issues/17

Phase 3 - Storage, dashboard, and Google Photos selection

#3 - Implement Cloud Storage lifecycle for source, output, and audio media [CLOSED / COMPLETE]

Owner: bhushannakhate | Area: Storage

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Configure Firebase project, environment variables, and service account access

Completed issue; scope, description, and dependencies intentionally unchanged in this revision.

github.com/bhushannakhate/project-2/issues/3

#1 - Implement image source upload and validation [OPEN]

Owner: bhushannakhate | Area: Storage / image validation

Dependencies: Implement Cloud Storage lifecycle for source, output, and audio media

github.com/bhushannakhate/project-2/issues/1

#18 - Build responsive dashboard and creation library [OPEN]

Owner: RyanTorrs | Area: Frontend

Dependencies: Implement creation CRUD API, Build the React sign-in and account experience

github.com/bhushannakhate/project-2/issues/18

#15 - Implement Google Photos OAuth 2.0 connection and secure token handling [OPEN]

Owner: bhushannakhate | Area: Google Photos OAuth

Dependencies: Configure Firebase project, environment variables, and service account access, Implement Firebase Authentication API and protected routes

github.com/bhushannakhate/project-2/issues/15

#19 - Build Google Photos Picker selection and source import [OPEN]

Owner: bhushannakhate | Area: Google Photos Picker

Dependencies: Implement Google Photos OAuth 2.0 connection and secure token handling, Implement image source upload and validation, Implement Cloud Storage lifecycle for source, output, and audio media

github.com/bhushannakhate/project-2/issues/19

Phase 4 - AI editing workflow

#5 - Create text and optional voice instruction editor [OPEN]

Owner: RyanTorrs | Area: Editing UI

Dependencies: Implement image source upload and validation, Implement Cloud Storage lifecycle for source, output, and audio media

github.com/bhushannakhate/project-2/issues/5

#24 - Implement Gemini Vision analysis and editing prompt preparation [OPEN]

Owner: RyanTorrs | Area: Gemini/Vision

Dependencies: Configure Firebase project, environment variables, and service account access, Implement image source upload and validation

github.com/bhushannakhate/project-2/issues/24

#12 - Build audio transcription endpoint and Cloud Storage integration [OPEN]

Owner: RyanTorrs | Area: Audio/AI

Dependencies: Create text and optional voice instruction editor, Implement Cloud Storage lifecycle for source, output, and audio media

github.com/bhushannakhate/project-2/issues/12

#8 - Implement Gemini image generation workflow with credits [OPEN]

Owner: RyanTorrs | Area: Gemini

Dependencies: Implement Gemini Vision analysis and editing prompt preparation, Create text and optional voice instruction editor, Implement authenticated user profile API and credits, Implement Cloud Storage lifecycle for source, output, and audio media

github.com/bhushannakhate/project-2/issues/8

#10 - Implement review and one-revision workflow [OPEN]

Owner: RyanTorrs | Area: Review

Dependencies: Implement Gemini image generation workflow with credits, Implement creation CRUD API

github.com/bhushannakhate/project-2/issues/10

#23 - Create multi-agent orchestration with Google ADK [OPEN]

Owner: bhushannakhate | Area: Agents

Dependencies: Implement Gemini Vision analysis and editing prompt preparation, Create text and optional voice instruction editor, Build audio transcription endpoint and Cloud Storage integration, Implement Gemini image generation workflow with credits, Implement review and one-revision workflow

github.com/bhushannakhate/project-2/issues/23

#26 - Optional: save approved creations to Google Photos [OPEN]

Owner: RyanTorrs | Area: Optional Google Photos export

Dependencies: Implement Google Photos OAuth 2.0 connection and secure token handling, Implement Gemini image generation workflow with credits, Implement review and one-revision workflow, Implement Cloud Storage lifecycle for source, output, and audio media

Optional; separate Library API export of a new app-created image. Not a core-editing or release dependency.

github.com/bhushannakhate/project-2/issues/26

Phase 5 - Quality, operations, and release

#20 - Add automated API, integration, and frontend tests [OPEN]

Owner: RyanTorrs | Area: Testing

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Implement creation CRUD API, Build the React sign-in and account experience

github.com/bhushannakhate/project-2/issues/20

#25 - Deploy backend and frontend to Google Cloud with public URLs [OPEN]

Owner: bhushannakhate | Area: Deployment

Dependencies: Implement Gemini image generation workflow with credits, Implement review and one-revision workflow, Build Google Photos Picker selection and source import, Implement Google Photos OAuth 2.0 connection and secure token handling, Configure CI, linting, formatting, and pull-request checks

Optional Google Photos export (#26) is explicitly not a release blocker.

github.com/bhushannakhate/project-2/issues/25

#6 - Add structured logging, metrics, alerts, and Cloud Run evidence [OPEN]

Owner: RyanTorrs | Area: Operations

Dependencies: Deploy backend and frontend to Google Cloud with public URLs

github.com/bhushannakhate/project-2/issues/6

#7 - End-to-end usability, accessibility, and responsive QA pass [OPEN]

Owner: bhushannakhate | Area: QA

Dependencies: Deploy backend and frontend to Google Cloud with public URLs

github.com/bhushannakhate/project-2/issues/7

#9 - Complete README, architecture documentation, setup guide, and GitHub Wiki [OPEN]

Owner: bhushannakhate | Area: Documentation

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Deploy backend and frontend to Google Cloud with public URLs

github.com/bhushannakhate/project-2/issues/9

#13 - Record rubric evidence: screenshots, database CRUD, commits, issues, and demo video [OPEN]

Owner: bhushannakhate | Area: Evidence

Dependencies: Deploy backend and frontend to Google Cloud with public URLs, Add structured logging, metrics, alerts, and Cloud Run evidence, Complete README, architecture documentation, setup guide, and GitHub Wiki

github.com/bhushannakhate/project-2/issues/13

Official Google references

Google Photos Picker: Get started — OAuth, session, Picker UI, polling, selected-item listing, and retrieval.

Google Photos Picker: Sessions — session lifecycle and polling configuration.

Google Photos Picker: Media items — retrieve selected media and temporary base URLs.

Google Photos authorization — minimum scopes, OAuth verification, and no service-account access.

Google Photos Library API — optional access to media created by the application.

Google Photos API user-data policy — purpose limitation, disclosure, consent, and Limited Use rules.

Stepwise handoff plan

Bhushan: continue the media and Google Photos integration slice (#1, #15, #19), coordinate with Ryan on API contracts, then own orchestration, deployment, documentation, and evidence.
Ryan: continue the editing UI, Vision/Gemini, review, and test work; coordinate with Bhushan on the Photos OAuth/import contract and user-consent boundary.
Keep Firebase Authentication and Google Photos OAuth as distinct credentials and UX states. Do not start generation until a valid source (#1/#19), credits (#4), editing instructions (#5), storage (#3), and Vision (#24) are ready. Optional Photos export (#26) can follow the core workflow and does not block deployment.

Sprint	Handoff	Issues	Division of work
1	Foundation and CI	#2, #14	Ryan completes backend foundation; coordinate CI and infrastructure conventions.
2	Firebase/data contract	#22, #21, #16	Ryan configures Firebase/auth; Bhushan defines Firestore and protected APIs.
3	User and CRUD vertical slice	#11, #4, #17	Bhushan owns UI profile/data integration; Ryan validates auth API behavior.
4	Media and Photos Picker	#3, #1, #15, #19	Bhushan owns storage/OAuth/import; Ryan reviews API contracts, UX states, and failures.
5	AI workflow	#5, #12, #24, #8, #10, #23	Ryan owns inputs/transcription/generation/review; Bhushan owns orchestration. Optional #26 Photos export is a separate follow-up.
6	Release	#20, #25, #6, #7, #9, #13	Bhushan owns deployment/docs/evidence; Ryan owns tests/logging/QA support.

Rubric grading map and evidence checklist

The point values below reproduce the readable values from the supplied rubric. Keep evidence current with the deployed application and approved Photos Picker flow.

Rubric criterion	Points	What earns full credit	Issue evidence
Documentation - application intro	5	Purpose, application name, public deployed URL and team context.	README, issue 9, and issue 13
Development description	5	Commit history, issue-board snapshots over time, major-file descriptions, development narrative.	Issues 13 and 14
Demonstration of application	10	Evidence of both team members using the app, Google Photos Picker selection/import, editing, and resulting application change.	Issues 13 and 19
Google Database/Firestore	20	Setup, structure, CRUD from app, queries, before/after data and retrieval evidence.	Issues 21, 17, and 13
Code styling and comments	20	Consistent naming and source-code clarity as required by the rubric.	Issues 2 and 14
Code repository completeness	5	All code, HTML, images, media and supporting files in GitHub.	All implementation issues
Analytics frontend and backend	15	Both Google Analytics integrations, useful trends and working implementation.	Issue 6 and issue 13
Analytics visuals and clarity	5	Clear charts/visuals suitable for explaining results.	Issue 13
Analytics interpretation	5	Comprehensive explanation of trends and what they mean for users/the app.	Issue 13
Analytics limitations	5	Accurate and detailed discussion of data, analysis, and conclusion limitations.	Issue 13
Cloud logging	10	Structured implementation and demonstration of calls/APIs in Cloud Logging.	Issue 6
Interface functionality	10	Secure Google Photos selection/import, task selection, loading/success/error feedback, and seamless core workflow.	Issues 5, 18, 19, 10, and 7
Interface style	10	Visual appeal, consistency, responsive layout and accessibility.	Issue 7
Fully functioning application	60	Reliability, completeness, Google Photos Picker and Gemini/Vision integration, review, and deployment.	Issues 8, 10, 19, 25, and 7
Firestore integration	40	Secure stored image metadata/results, unique IDs/users, retrieval and CRUD.	Issues 21, 17, 3, and 8
Vision and Gemini integration	points vary	Meaningful Vision advanced feature, Gemini prompting/results, error handling and documentation.	Issues 24 and 8
Google Cloud deployment	40	Public Cloud Run/GAE deployment with stable startup, secure configuration, scaling and working features.	Issue 25
YouTube video and final summary	30 stated or source rendering unclear; verify Canvas	Ordered introduction, proposal, GitHub, app demo, PaaS console, database, and summary sections.	Issue 13

Priority grading strategy

First protect high-value points: (1) public Cloud deployment, (2) fully functioning application, (3) Firestore database/CRUD, (4) Vision/Gemini, (5) interface style/function, and (6) logging.
Then capture evidence: show the deployed app, database before/after changes, issue-board history, commits, Cloud console/logging, both teammates, user consent, Google Photos Picker selection/import, generated result, and failure recovery.
Pivot mapping: the former Pinterest selection/publishing references are replaced by Google Photos Picker selection/import for the core source-photo requirement. Photos export is optional and separately scoped. Confirm with the instructor if the rubric requires a particular named provider.